University of Canterbury

EMTH210

ENGINEERING MATHEMATICS

Topic 2: Differential equations

2.1 SECOND ORDER HOMOGENEOUS LINEAR CONSTANT COEFFICIENT ODEs

These are things of the form:

$$a_2 \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + a_1 \frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = 0,$$

where a_2, a_1 and a_0 are constants with $a_2 \neq 0$.

Second order because ____.

Homogeneous because ____.

Linear because ____.

Constant coefficient because ____.

We saw last year that DEs like this have solutions of the form $y = e^{mt}$, where m is constant (and possibly complex). When we know the form of the solution but not its details (e.g. the value of m), the form is called an *ansatz*. So let's try the ansatz $y = e^{mt}$ in the 2nd order ODE.

Example 2.1.1

Solve the following DE:

$$y'' + 3y' + 2y = 0.$$

Solution.

♦

The linearity of the equation lets us take a linear combination of the two component solutions to get the general solution. So this method will not work for non-linear equations!

We should check that our general solution does indeed solve the DE:

$$\begin{aligned}
\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 3\frac{\mathrm{d}y}{\mathrm{d}t} + 2y &= \frac{\mathrm{d}^2}{\mathrm{d}t^2}(C_1 e^{-t} + C_2 e^{-2t}) + 3\frac{\mathrm{d}}{\mathrm{d}t}(C_1 e^{-t} + C_2 e^{-2t}) + 2(C_1 e^{-t} + C_2 e^{-2t}) \\
&= C_1 \left(\frac{\mathrm{d}^2}{\mathrm{d}t^2} e^{-t} + 3\frac{\mathrm{d}}{\mathrm{d}t} e^{-t} + 2e^{-t} \right) + C_2 \left(\frac{\mathrm{d}^2}{\mathrm{d}t^2} e^{-2t} + 3\frac{\mathrm{d}}{\mathrm{d}t} e^{-2t} + 2e^{-2t} \right) \\
&= C_1 \cdot 0 + C_2 \cdot 0 = 0.
\end{aligned}$$

The steps for solving a second order homogeneous linear constant coefficient ODE

$$a_2y'' + a_1y' + a_0y = 0$$

are:

1. Ensure the ODE is 2nd order, homogeneous, linear, constant coefficients.
2. Create the *ansatz* $y = e^{mt}$ (or $y = e^{mx}$, etc., depending on the variable names).
3. Substitute the ansatz into the ODE.
4. Obtain and solve the *auxiliary equation*, which will be a second order **algebraic** equation like $a_2m^2 + a_1m + a_0 = 0$, to get two values m_1, m_2.
5. Write down the *general solution* of the ODE, which is one of:
   - $y(t) = C_1\, e^{m_1\, t} + C_2\, e^{m_2\, t}$ (distinct real roots $m_1 \neq m_2$)
   - $y(t) = C_1\, t\, e^{m_1\, t} + C_2\, e^{m_1\, t}$ (repeated roots $m_1 = m_2$)
   - $y(t) = e^{at}\,(C_1\, \cos bt + C_2\, \sin bt)$ (complex conjugate pair of roots $m_2 = \overline{m}_1$, where $a = \mathrm{Re}(m_1)$ and $b = \mathrm{Im}(m_1)$, so $m_1 = a + i\, b$ and $m_2 = a - i\, b$)

   where C_1, C_2 are arbitrary real constants.
6. If you have any extra information (initial or boundary conditions, say) then use these to find the constants C_1, C_2.

Using the discriminant, you can look at a second order homogeneous linear constant coefficient ODE and quickly say something about how the solutions will behave before you solve the ODE. This can be tremendously useful.

Note that the solution of a second order has two unknowns which can be found with two conditions. If the conditions are given at the start, this is called an *initial value problem*, or *IVP*.

Example 2.1.2

Find C_1 and C_2 for the previous example if the initial conditions are

$$y(0) = 1, \quad y'(0) = 3.$$

Solution.

♦

The power of this method of solution lies in reducing the problem of solving an ODE to the problem of solving an algebraic equation, namely the auxiliary equation $am^2+bm+c = 0$. Recall that when the discriminant $b^2 - 4ac$ of this quadratic equation is positive, zero, or negative, we get two distinct real roots, one repeated real root, or two complex conjugate roots, respectively.

Example 2.1.3

Solve

$$y'' + 6y' + 13y = 0.$$

Solution.

♦

Two comments.

1. The general solution is real. Here, that link between complex numbers and trigonometric functions becomes essential to a very real application. Of course a physical situation modelled by this ODE would have to have real, not complex, solutions. So even though the method involves complex numbers, the solution is real.

2. From now on *do not* go through the rigmarole of Euler's formula etc. — just take the shortcut and **go directly** from $m = -3 \pm 2i$ to

$$y = e^{-3t}\left[C_1 \cos(2t) + C_2 \sin(2t)\right].$$

At this point, we can glance at the coefficients of an ODE, imagine they were the coefficients of the auxiliary equation, and if the discriminant thereof is positive we can predict that the solution will involve exponential growth, decay, or a combination of the two, while if the discriminant is negative we can predict the existence of cyclic solutions (combinations of sines and cosines). Next, we see what happens when the discriminant is zero.

2.2 REPEATED ROOTS

This is what we call the situation when the discriminant of the auxiliary equation is zero, as in this example:

$$y'' + 2y' + y = 0.$$

As usual, we put $y = e^{mt}$, and obtain the auxiliary equation

$$\begin{aligned} m^2 + 2m + 1 &= 0 \\ (m+1)^2 &= 0 \end{aligned}$$

so $m = -1, -1$. This gives only ONE linearly independent component solution $y_1 = e^{-t}$. But we expect a second component solution as the ODE is second order. The general rule for repeated roots is that $\boxed{\text{if } y_1 = e^{m_1 t} \text{ is the first component solution then the second}}$

$\boxed{\text{component solution is simply } y_2 = te^{m_1 t}}$, that is, t times the first solution. For the current example, we can see that this works:

$$\begin{aligned} & y_2(t) = te^{-t}, \\ \implies & y_2'(t) = (1)e^{-t} + t(-e^{-t}) = (1-t)e^{-t} \\ \implies & y_2''(t) = (-1)e^{-t} + (1-t)(-e^{-t}) = (-2+t)e^{-t}, \end{aligned}$$

and so

$$y_2'' + 2y_2' + y_2 = (-2+t)e^{-t} + 2(1-t)e^{-t} + te^{-t} = 0.$$

So as y_2 satisfies the original DE we know that this is the second solution. Therefore the general solution for this example is

$$y = C_1 e^{-t} + C_2 te^{-t},$$

where C_1 and C_2 are arbitrary constants.

In general, (i.e. for higher roots), if a root $m = \lambda$ occurs k times then a DE has k linearly independent component solutions,

$$e^{\lambda t},\ te^{\lambda t},\ t^2 e^{\lambda t},\ \ldots,\ t^{k-1} e^{\lambda t},$$

plus solutions from any other roots.

We have now seen the three types of solution possible for second order homogeneous linear constant coefficient ODEs, and how they occur depending on the roots of the auxiliary equation. We now move on to finding values of the constants C_1, C_2 depending on extra information with which we may be supplied.

2.3 BOUNDARY VALUE PROBLEMS (BVPs)

These differ from initial value problems in that the extra information involves *more than one* location (usually position x, rather than time t).

Example 2.3.1

Solve the boundary value problem:

$$y'' + \omega^2 y = 0$$

$$y(0) = 0 \quad \text{and} \quad y(L) = 0 \quad (L \neq 0)$$

Solution.

♦

There is another possibility that can occur with different boundary conditions, as we see in the next example.

Example 2.3.2

As before,

$$y'' + \omega^2 y = 0,$$

but now

$$y(0) = 0 \quad \text{and} \quad y(L) = 1, \quad L \neq 0$$

Solve the new boundary value problem.

Solution.

♦

So in both examples, we needed to think about the zeroes of the function $\sin(x)$. In the first example, this was to identify the two possible situations: the trivial solution, or an infinite family of (non-trivial) solutions. In the second example, the two possible situations were: a unique solution, or no solutions. In both cases, the number and type of solution depends on the value of ωL. You should think about what this means.

In contrast, and you should check this, an initial value problem $y'' + \omega^2 y = 0$ with $y(t_0) = y_0$ and $y'(t_0) = y_0'$ *always* has a unique solution.

2.4 APPLICATION: BUCKLING OF A THIN BEAM

Sketch of the situation:

This is not a time-dependent problem, but rather $y = y(x)$ where y is the downwards displacement of the beam at horizontal position x. The beam is pinned freely at the ends, so that $y(0) = y(L) = 0$ but $y'(0)$ and $y'(L)$ are free. The beam is under compression by a force P.

The governing equation and boundary conditions of the beam are

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{P}{EI} y = 0 \quad , \quad y(0) = y(L) = 0,$$

where E is Young's modulus and I is the second moment of the cross-sectional area of the beam[1].

This has general solution

[1]The derivation is as follows. If $M(x)$ is the moment on the left-hand part of the beam exerted by the right-hand part of the beam at a point C, then taking moments about C gives

$$Py(x) + M(x) = 0.$$

For small displacements

$$M(x) = EI \frac{\mathrm{d}^2 y}{\mathrm{d}x^2}.$$

Substituting the latter into the former gives the governing equation.

Then the boundary conditions give:

so that either $C_2 = 0$, in which case we have the trivial solution $y = 0$ (beam stays straight), or $kL = n\pi$ for some integer n, in which case we have the solution

$$y = C_2 \sin\left(\frac{n\pi x}{L}\right),$$

where C_2 is arbitrary. In particular, C_2 can be arbitrarily large — the beam buckles.

We may now wish to ask under which conditions the beam will buckle. Well,

Hence the beam buckles for these values of P, which are known as *critical loads*. Note that these are steady-state results, not dynamic, so some care is required when interpreting these results. The smallest force P to cause buckling is

which is the so-called "Euler load".

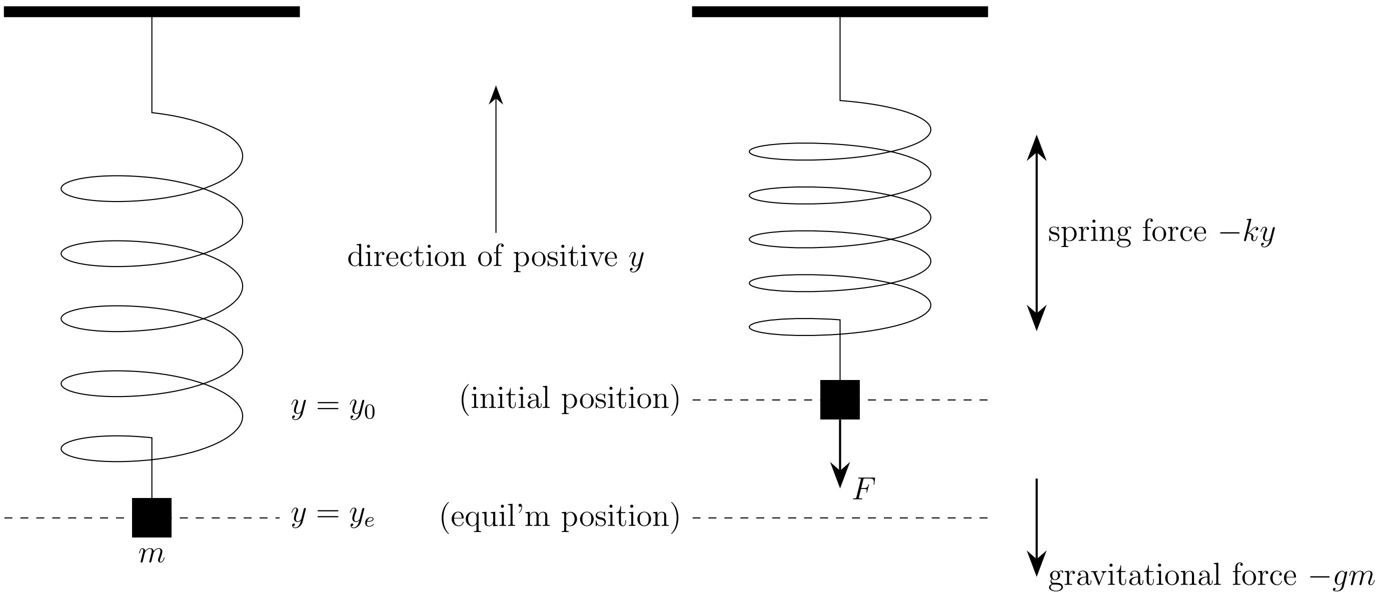

[2]

Good old Euler; just look at that hat.

[2]Portrait by Jakob Emanuel Handmann (1718-1781). File downloaded from commons.wikimedia.org, Public Domain.

2.5 INHOMOGENEOUS LINEAR ODEs

Motivating problem: *describe mathematically the motion of a weight on a spring.*



From laws of physics

$$m\,y'' = F_{total} = F_{gravity} + F_{spring} + F_{damping} = -g\,m \,-\, k\,y \,-\, \rho y'.$$

The DE is then

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p\,\frac{\mathrm{d}y}{\mathrm{d}t} + q\,y = -g,$$

where $p = \dfrac{\rho}{m}$ and $q = \dfrac{k}{m}$. A particular solution is $y_p = y_e = \dfrac{-mg}{k}$, the equilibrium displacement for the spring mass system.

We could also add other external forces to the right-hand side (which can depend on t).

Mathematical problem: *how do we solve this differential equation?*

In general, inhomogeneous linear second-order DEs with constant coefficients are DEs of the form

$$ay'' + by' + cy = f(t), \tag{1}$$

where you will note that the right-hand side (RHS) can now be non-zero.

The inhomogeneous linear ODE (1) has a general solution $y(t) = y_p(t) + y_c(t)$ where $y_p(t)$ is *any* solution to (1) and $y_c(t)$ is the general solution to

$$a_2 y'' + a_1 y' + a_0 y = 0.$$

This works because

$$a_2(y_p'' + y_c'') + a_1(y_p' + y_c') + a_0(y_p + y_c) = \underbrace{a_2 y_p'' + a_1 y_p' + a_0 y_p}_{f(t)} + \underbrace{a_2 y_c'' + a_1 y_c' + a_0 y_c}_{0} = f(t).$$

Terminology:

- $y_p(t)$ is called the *particular integral* (PI);
- $y_c(t)$ is called the *complementary function* (CF).
  (Also called $y_h(t)$, the homogeneous solution.)

The general strategy for solving an inhomogeneous linear ODE is

1. Set the RHS equal to zero, and solve the resulting homogeneous ODE (by the methods we have learned) to find the complementary function $y_c(t)$.
2. Find any solution to the full inhomogeneous equation (by the methods to be outlined below) — this is the particular integral $y_p(t)$.
3. The general solution is $y(t) = y_p(t) + y_c(t)$.

Before discussing the general method for finding y_p in §2.6, let's see the ideas in action.

Example 2.5.1

Find the general solution of

$$y'' + 3y' + 2y = 4t + 8.$$

Solution.

♦

This method of "guessing" the form of the RHS but with unknown constant is known as the *method of undetermined coefficients*, because the method involves coefficients (A and B in the previous example) which are undetermined[3]

[3] And because "big fat guess" doesn't sound impressive enough.

2.6 BASIC STRATEGY FOR "GUESSING" FORM OF y_p — THE *METHOD OF UNDETERMINED COEFFICIENTS*

The following four rules systematically generate every possible term that could come from the RHS and its derivatives, all with (to start with) undetermined coefficients. The *form* of the y_p generated by the four rules is the ansatz for y_p, which has the coefficients determined.

Follow the four rules carefully and *in order* and you won't go wrong. When we say "in order" we really mean it: do not apply rule 3, then rule 4, then go back to rule 2. Do 1, then 2, then 3, then 4, and never look back.

1. **Use undetermined coefficients**

   Set y_p to the RHS, with the coefficients replaced by undetermined coefficients A, B, etc.

   E.g. $7x^2 + \sin(2x) + xe^{3x} \longrightarrow Ax^2 + B\sin(2x) + Cxe^{3x}$.

2. **Fill in lower powers**

   Fill in lower polynomial powers all the way to constants.

   E.g. $Ax^2 + Cxe^{3x} \longrightarrow Ax^2 + Bx + C + Dxe^{3x} + Ee^{3x}$.

3. **Match** $\sin$ **and** $\cos$

   Put in matching sin terms for every cos, and vice versa.

   E.g. $A\sin(2x) + Be^x\cos(3x)$
   $$\longrightarrow A\sin(2x) + B\cos(2x) + Ce^x\sin(3x) + De^x\cos(3x).$$

4. **Prevent linearly dependent (overlapping) terms**

   For terms which are linearly dependent on other terms in y_c or y_p, multiply them by x until they are linearly independent.

   E.g. if $y_c = C_1e^x + C_2e^{-x}$, $Axe^x + Be^x \longrightarrow Axe^x + Bx^2e^x$.

   The form must make it impossible for terms of y_p to combine with each other or with terms in y_c when we add them with $y = y_p + y_c$.

Always explicitly label the relevant steps "Rule 1" and so on.

Example 2.6.1

Find the corresponding homogeneous solutions and the **form** of y_p for the following DEs.

(a) $y'' + 7y' - 8y = 7e^{-x}$

(b) $y'' + 7y' - 8y = 7e^{x}$

(c) $y'' + 4y' + 4y = e^{x} \sin x$

(d) $y'' + 4y' + 4y = 4\,x^2\,e^{-2x}$

(e) $y'' + y = e^{x}\ \sin x$

(f) $y'' + y = \sin x$

2.7 FORCED OSCILLATIONS

Interesting scenarios result when the system is subject to *external forcing*: for example, a mechanical actuator could lift and drop the top of the attachment point of a spring to the ceiling. This force can be inserted as a RHS to the equation, making it non-homogeneous:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p\,\frac{\mathrm{d}y}{\mathrm{d}t} + q\,y(t) = f(t).$$

A simple scenario (in the absence of damping) is the presence of a sinsusoidal forcing term, which induces interesting behaviour.

When the forcing frequency is equal to the natural frequency of a freely oscillating system, *resonance* occurs and the amplitude grows (without bound in these simple models; the real world obviously limits growth).

When the forcing frequency is close to the natural frequency of a freely oscillating system, interesting patterns can emerge (*beats*).

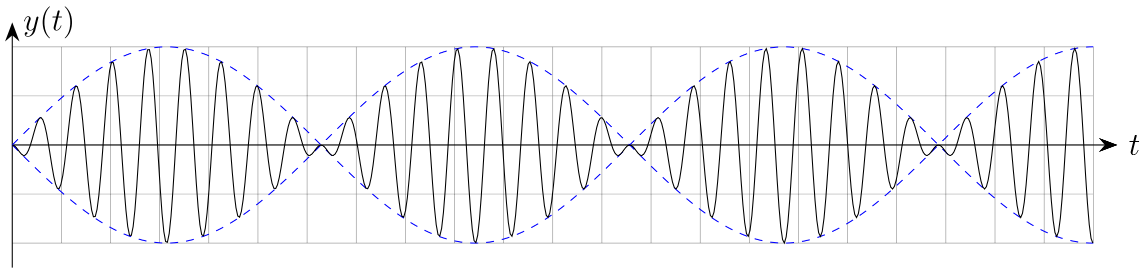
Example 2.7.1

Solve the initial value problem

$$y'' \;+\; 64\,y \;=\; -17\cos 9t\,; \qquad y(0) \;=\; 0,\; y'(0) \;=\; 0\,.$$

Solution.

♦



If in doubt about applying the method of undetermined coefficients, it is safer to include extra terms. If these terms are not needed the method will automatically set their unknown coefficients to zero except for terms which overlap with y_c; the coefficients of these terms will be arbitrary. On the other hand if necessary terms are omitted then the method will almost always fail.

Now we look at an example with resonance:

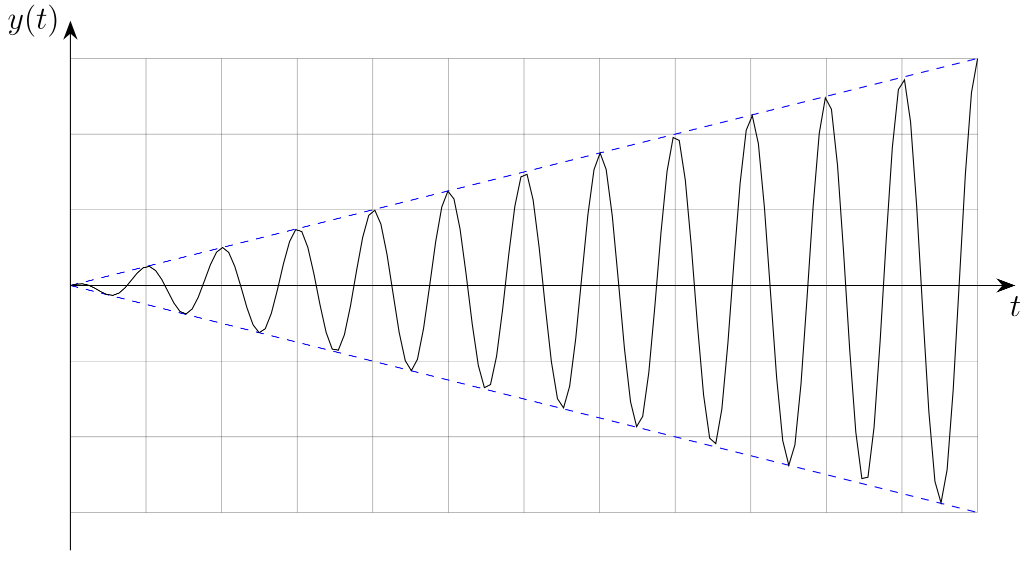
Example 2.7.2

Solve the initial value problem

$$y'' + 64\,y = -16\sin 8t, \quad \text{with} \quad y(0) = 0,\ y'(0) = 1\,.$$

Solution.

♦



$y(t)$
t

2.8 MORE EXAMPLES

Example 2.8.1

Find the general solution of

$$y'' - 2y' + y = t^2 e^t + 5.$$

Solution.

♦

Example 2.8.2

Find the general solution of

$$y'' + y = e^t \sin(t).$$

Solution.

♦

Example 2.8.3

Find the general solution of

$$y'' - y = te^t.$$

Solution.

♦

Example 2.8.4

Find the general solution of

$$y'' + y = \sin(t) + 3te^{-t}.$$

Solution.

♦

Example 2.8.5

Find the general solution of

$$\frac{d^3y}{dt^3} + \frac{\mathrm{d}y}{\mathrm{d}t} = t^2 + e^{2t}.$$

Solution.

♦

Note: the method of undetermined coefficients only really works for constant coefficient DEs when the RHS of $f(t)$ consists of terms with are polynomial, exponential, sine, cosine, or products thereof.